Lecture 6

Crystal Field Theory

Some model questions for major exam

Q1. Crystal field stabilization energy $[Cr(H_2O)_6]^{2+}$ is:

a) $– 1.8\ \Delta_0$
b) $– 1.6\ \Delta_0 + P$
c) $–1.2\ \Delta_0$
d) $–0.6\ \Delta_0$

Q2. State True or False.

Strong field ligands, such as $CN^–$ usually produce low spin complexes and large crystal field splitting.

Q3. Predict the shape of $[ZnCl_4]^{2-}$

Q4. Find out the magnetic moment of $[FeF_6]^{3-}$

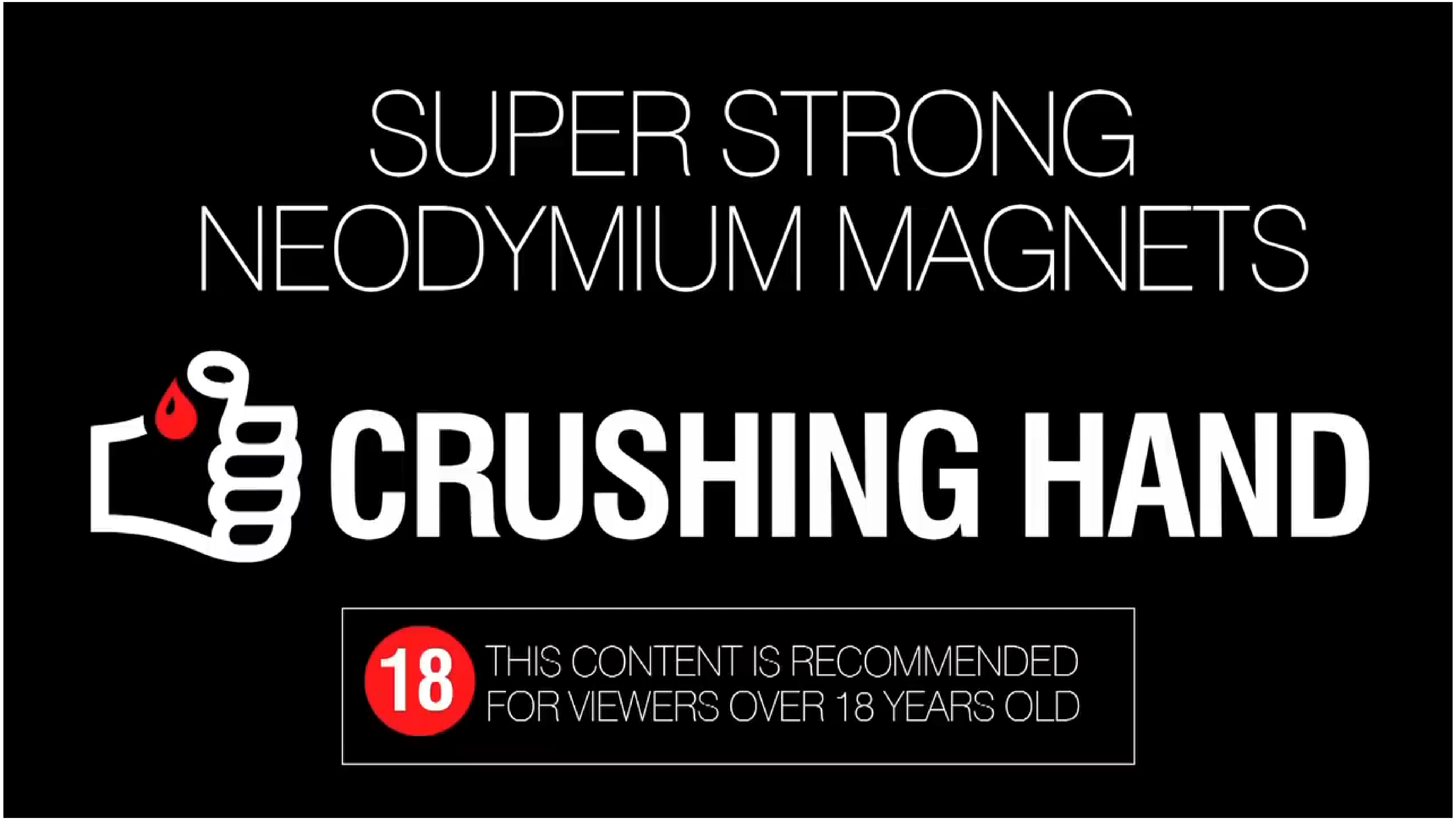
a) 1.73 BM
b) 2.83 BM
c) 3.87 BM
d) 5.92 BM

Problem solving !!

Cobalt form three octahedral complexes, $[MCl_6]^{4-}$, $[M(R_2NCS_2)_3]$ and $[M(CN)_5(H_2O)]^{3-}$. These have magnetic moments 0, 1.73 and 3.9 BM (not in the order). Assign the magnetic moments to the complexes and predict the relative strength of the dithiocarbamate ligand.

Solution:

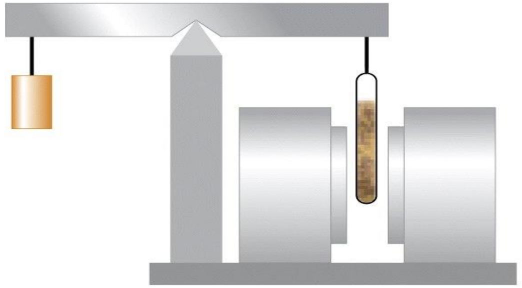
$[CoCl_6]^{4-}$ Co^{2+}, 3d ^{7},Cl- weak ligand, HS, 3 unpaired e's ~ 3.9 BM
$[Co(CN)_5(H_2O)]^{3-}$. Co^{2+}, CN- strong ligand, LS 1 unpaired e 1.73 BM
$[Co(R_2NCS_2)_3]$, Co^{3+}, 3d ^{6} , 0 BM means ligand is strong field and low spin complex



https://www.youtube.com/watch?v=0t8yDnyOaQ8&ab_channel=MagnetExpert

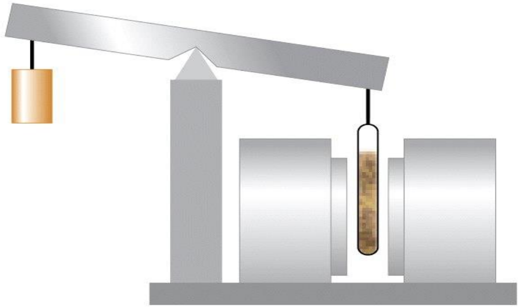
Magnetic properties of transition metal complexes

Magnet off

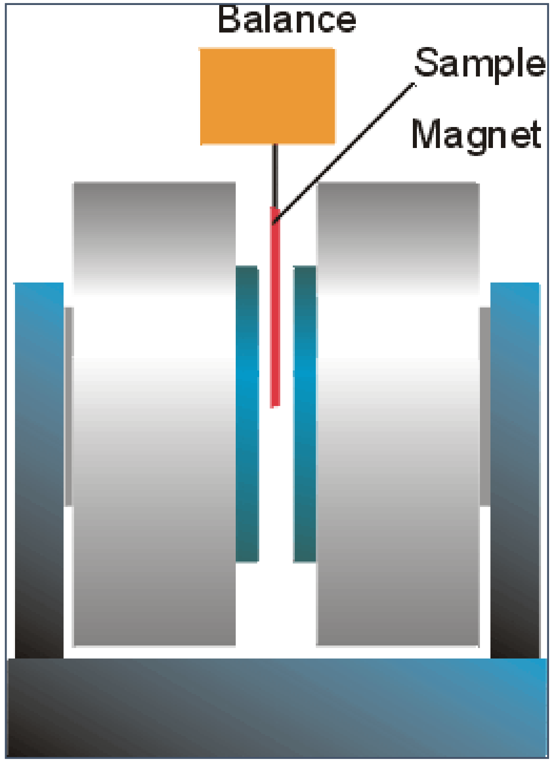


(a)

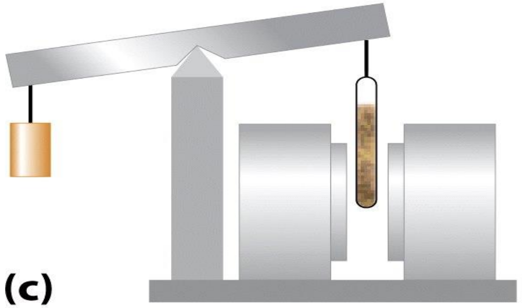
Magnet on:
Paramagnetic



(b)



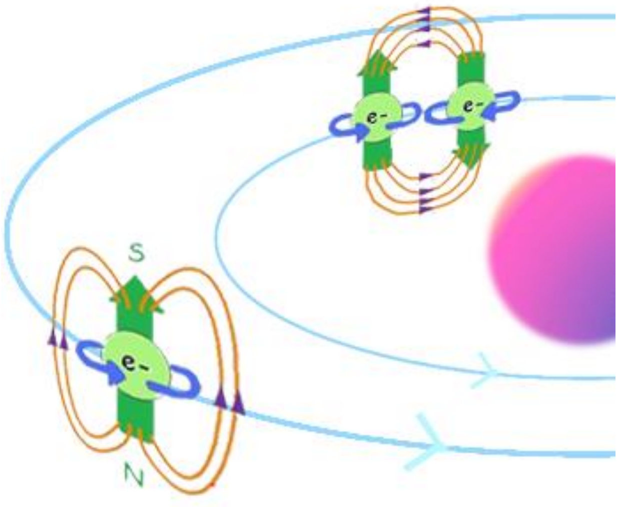
Magnet on:
diamagnetic



(c)

Gouy balance to measure the magnetic susceptibilities

Magnetic properties of metal complexes



Magnetism is caused by moving charged electrical particles (Faraday, 1830s). These particles can be the current of electrons through an electric wire, or the movement of charged particles (protons and electrons) within an atom. These charged particles move much like planets in a solar system:

nucleus spin around its own axis, causing a **very weak** magnetic field.
electrons orbit around the nucleus, causing a **weak** magnetic field.
electrons spin around their own axis, causing a **significant** magnetic field .

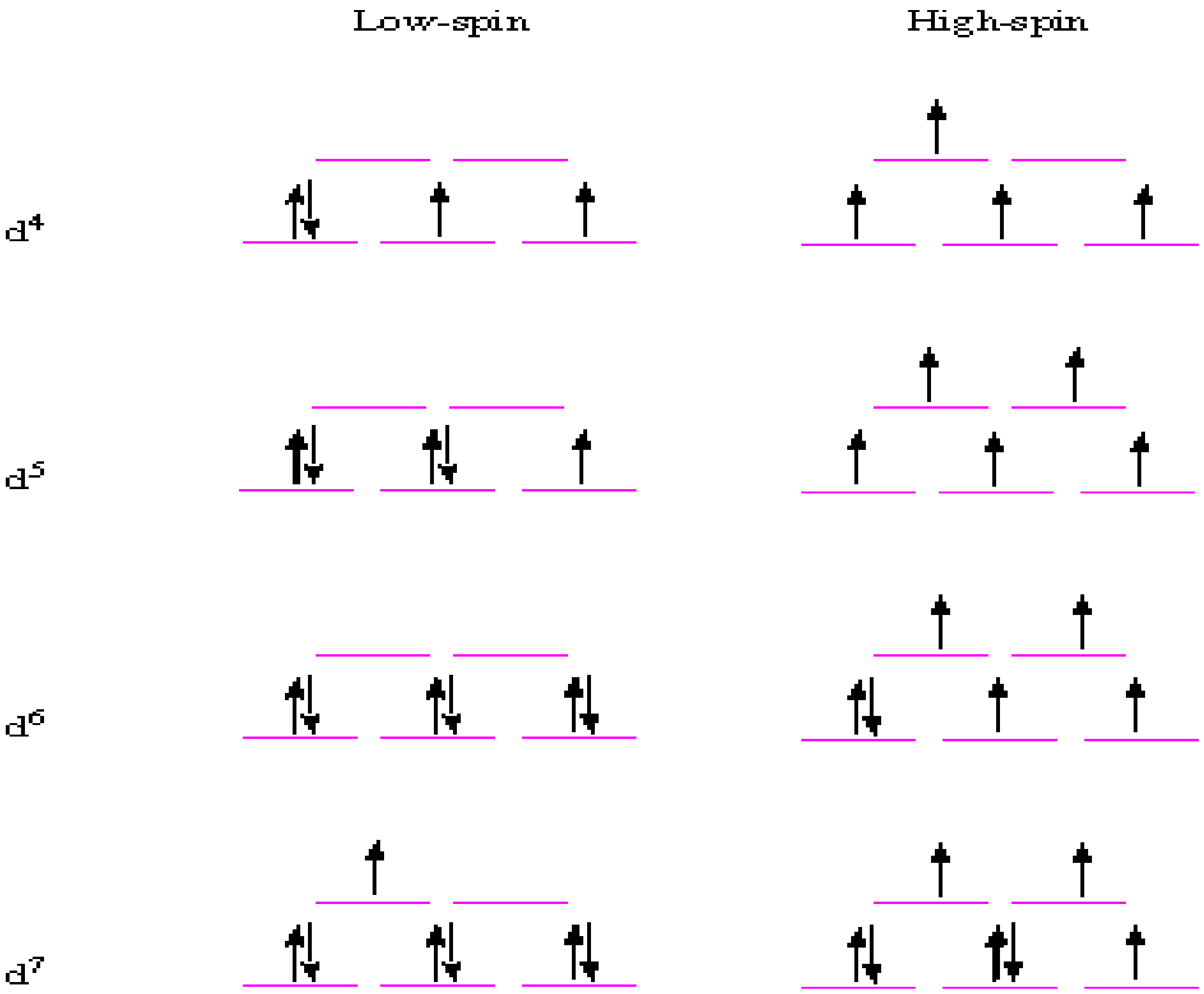
Spinning electrons generate the bulk of the magnetism in an atom.

Within each orbit, electrons with opposite spins pair together, resulting in no net magnetic field. Therefore **only unpaired electrons** lead to magnetic moment

The **spin-only** formula (μ_s)

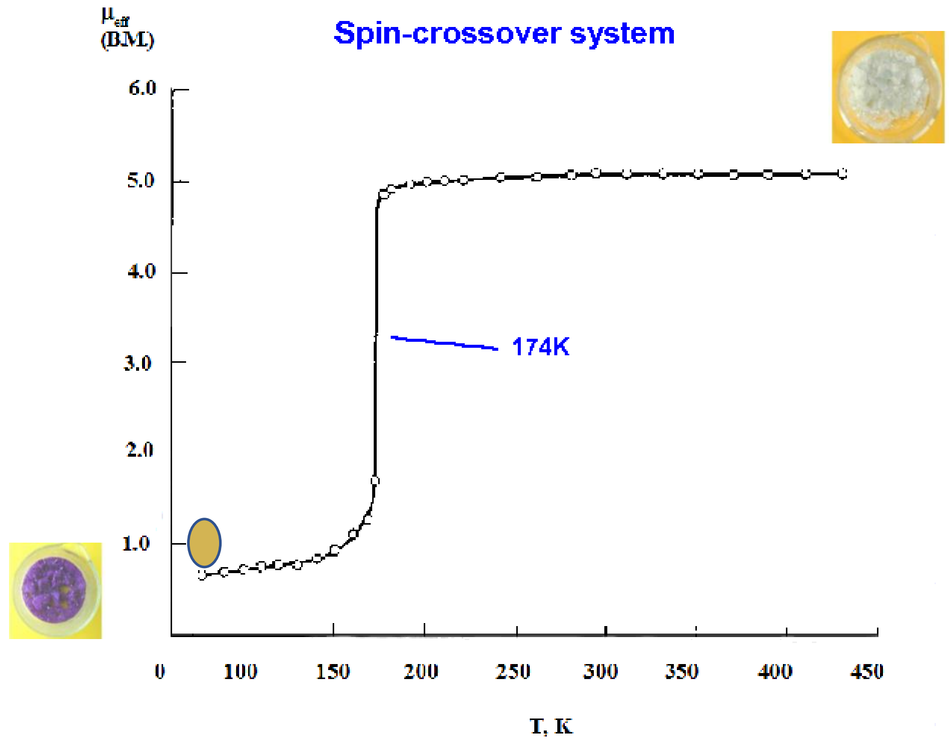
$$\boxed{\mu_s = \sqrt{n(n+2)}}$$

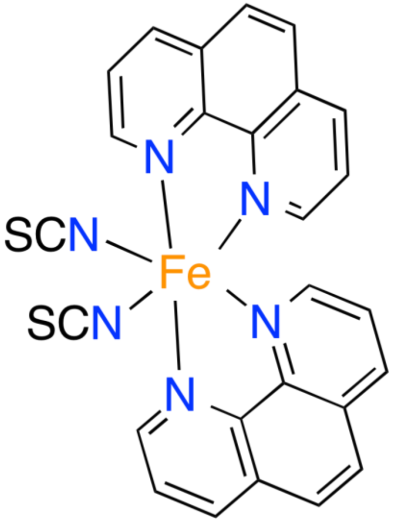
How Crystal Field theory explains magnetic moments of TM complexes to a decent extent?



Strength of the ligand, type of complex , size of the metal and its oxidation state affects crystal field splitting

Temperature dependence of magnetic moments



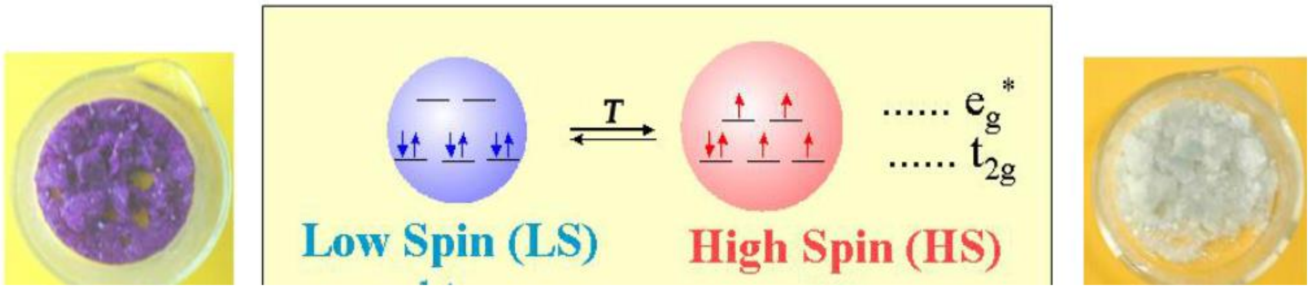


$[Fe(phen)_2(NCS)_2]$

Octahedral metal complexes with a d^4–d^7 electronic configuration can, in principle, exist in the high spin (HS) or low spin (LS) states. Such systems can be switched between the two spin states by using external perturbations such as temperature; this is a phenomenon that is known as spin crossover (SCO).

How crystal field theory explains temperature dependence of μ_{eff}

$Fe^{2+} = d^6$



$T_{1/2}$ is the temperature for which there is coexistence of 50% of LS and 50% of HS molecules.

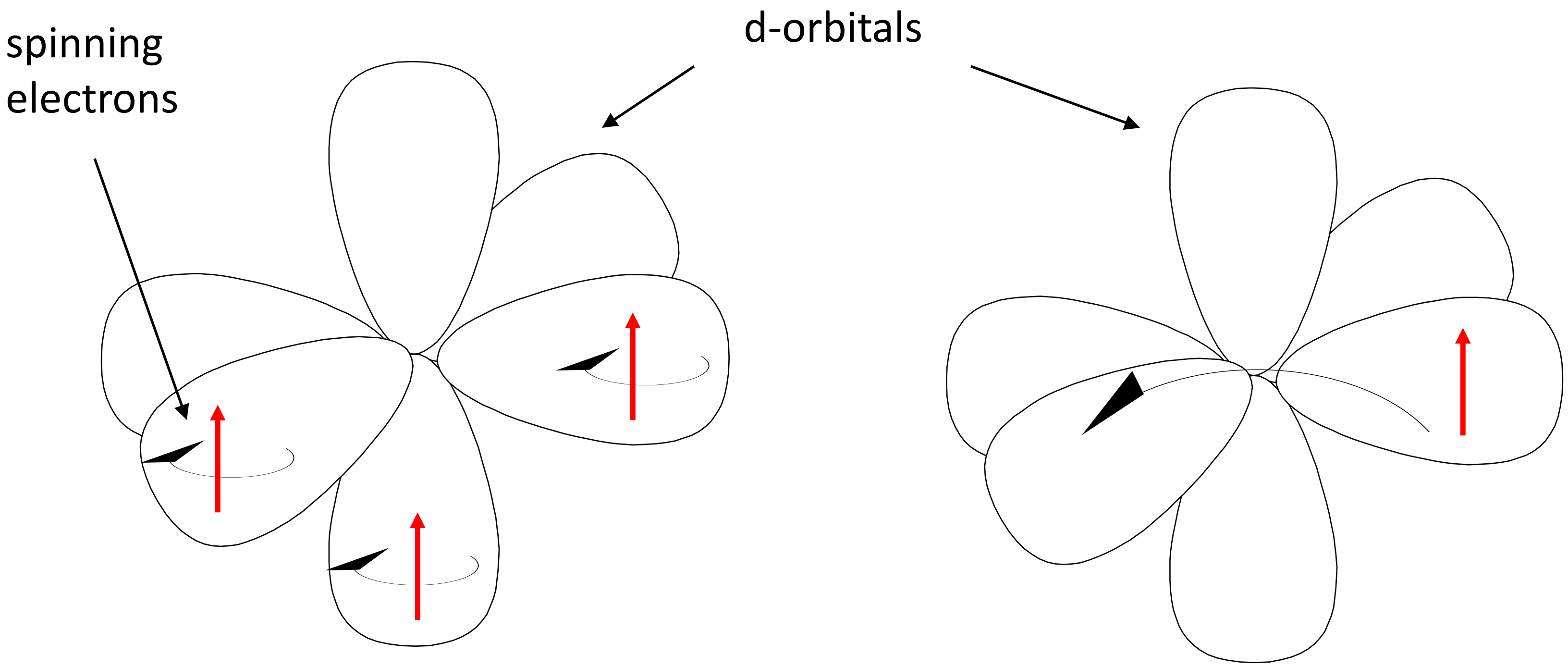
Magnetic properties: Spin only and effective

The **spin-only** formula (μ_s) applies reasonably well to metal ions from the first row of transition metals: (units = μ_B, Bohr-magnetons); magnetic moments value of high-spin complexes of first row d-block ions are given below.

Metal ion	d^n configuration	μ_s(calculated)	μ_{eff} (observed)
Ca^{2+}, Sc^{3+}	d^0	0	0
Ti^{3+}	d^1	1.73	1.7-1.8
V^{3+}	d^2	2.83	2.8-3.1
V^{2+}, Cr^{3+}	d^3	3.87	3.7-3.9
Cu^{2+}	d^9	1.73	1.9-2.1 (tetrahedral)
Zn^{2+}, Ga^{3+}	d^{10}	0	0

Spin and orbital contributions to μ_{eff}

For the first-row d-block metal ions the main contribution to magnetic susceptibility is from electron spin. However, there is also an orbital contribution (especially for the second and third row TM) from the motion of unpaired electrons from one d-orbital to another. This motion constitutes an electric current, and so creates a magnetic field.

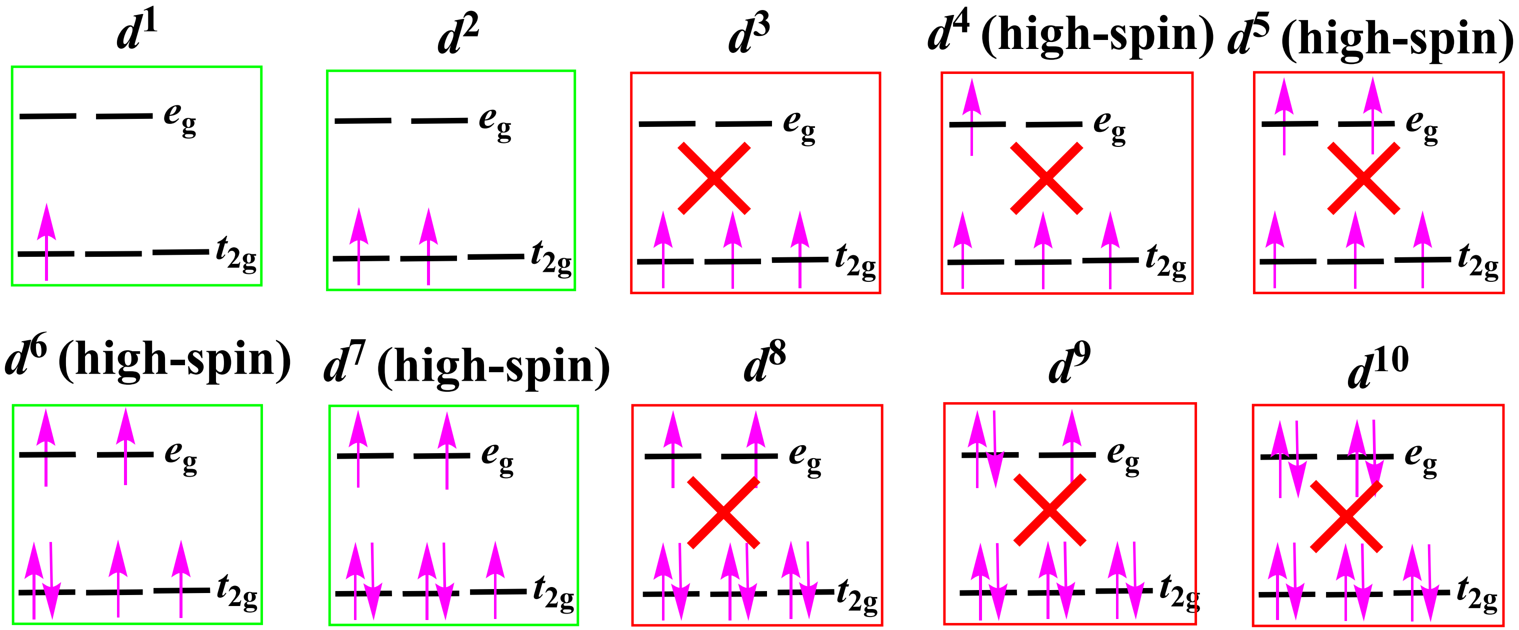


spin contribution – electrons are spinning creating an electric current and hence a magnetic field

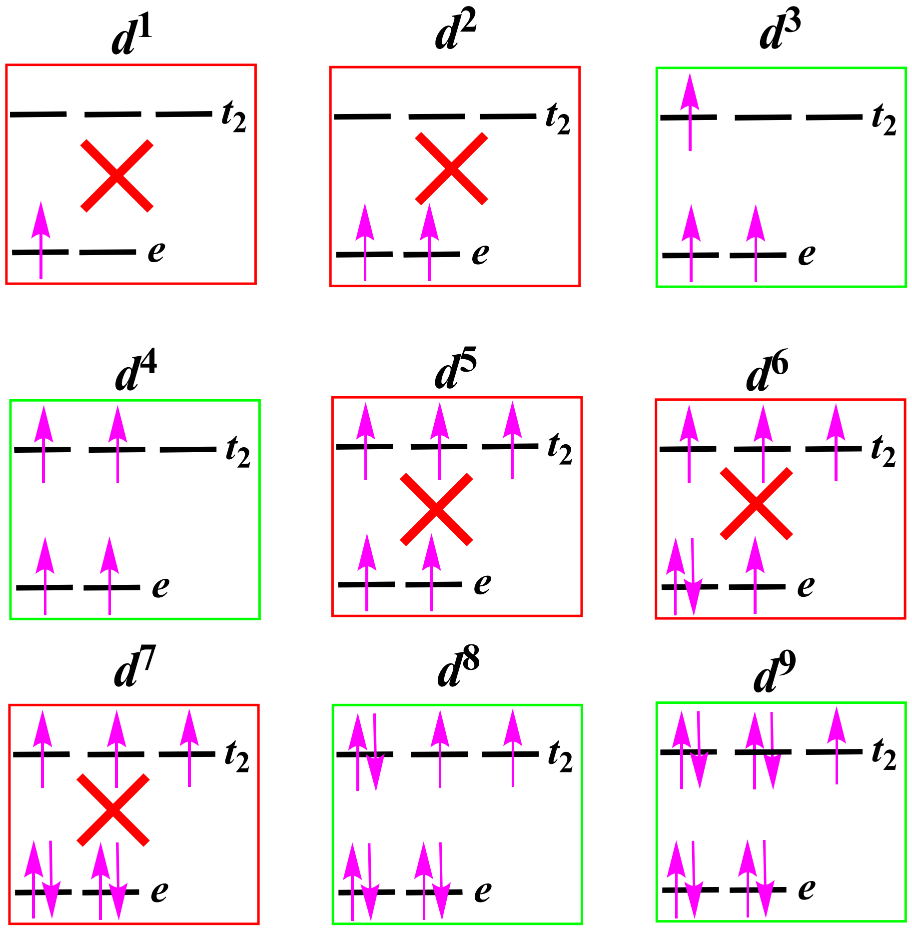
orbital contribution - electrons move from one orbital to another creating a current and hence a magnetic field

Orbital contribution to magnetic moment

Octahedral Complexes



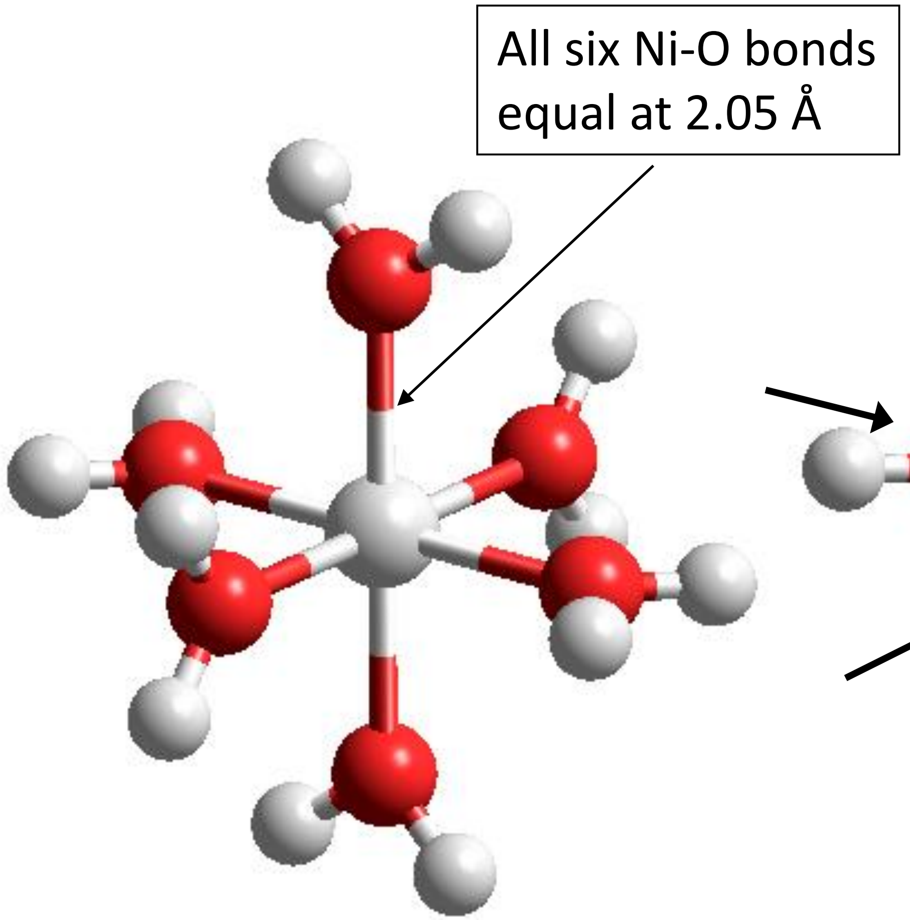
Tetrahedral Complexes



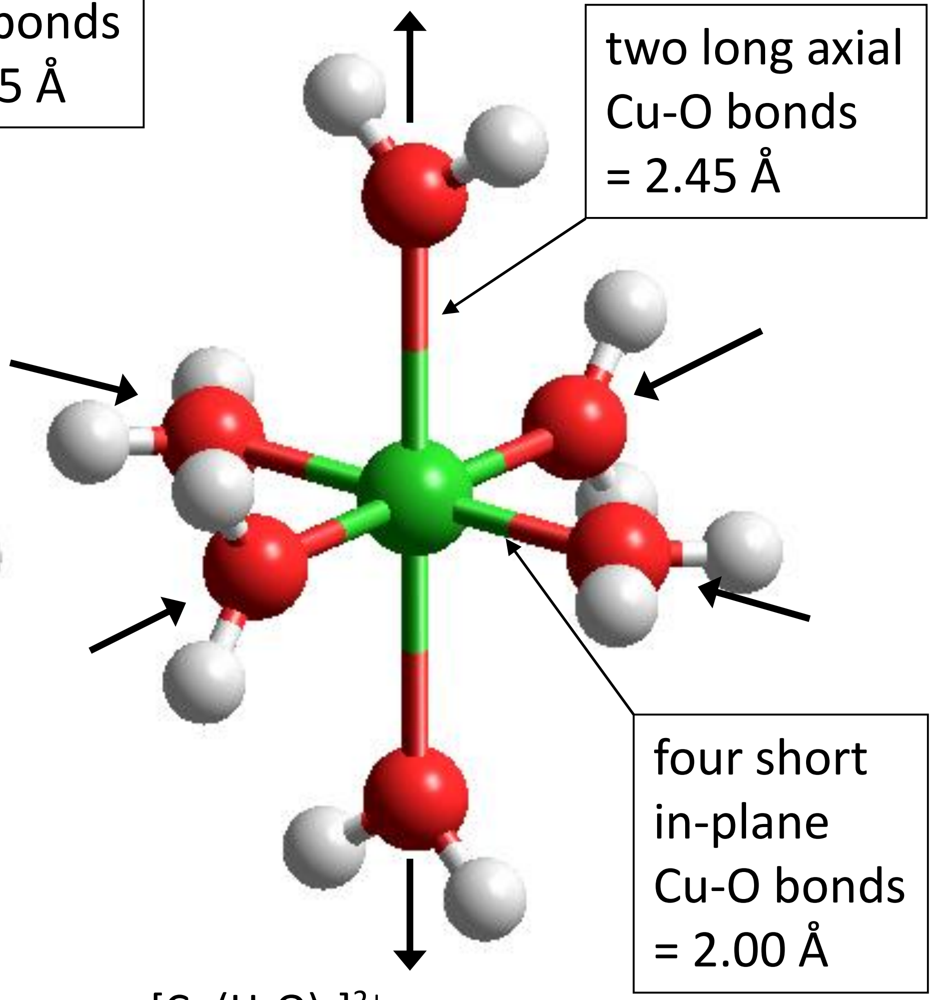
The potassium salt of $[Fe(CN)_6]^{3-}$ has $\mu = 2.3\mu_B$, which is between the spin-only values for one and two unpaired electrons (1.7 μ_B and 2.8 μ_B, respectively). Explain.

In this case, the spin-only assumption has failed because the orbital contribution to the magnetic moment is substantial.

Distortion of some octahedral complexes

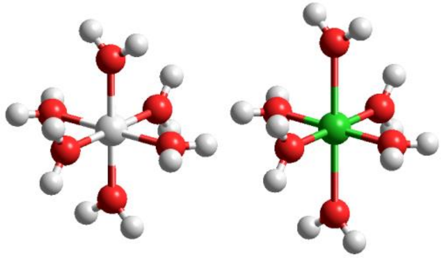


$[Ni(H_2O)_6]^{2+}$
no distortion



$[Cu(H_2O)_6]^{2+}$
distortion lengthens axial Cu-O's

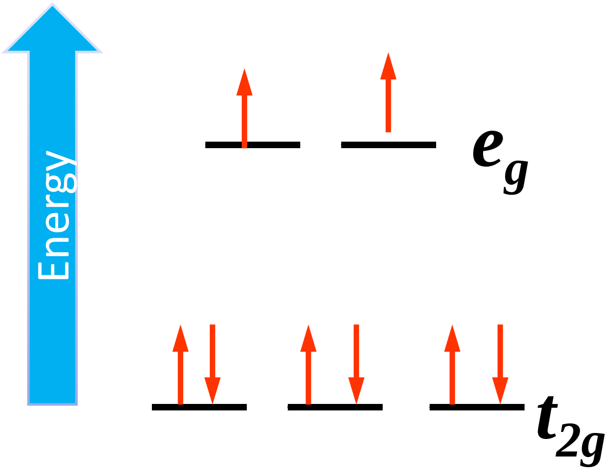
Ni(II) Cu(II)



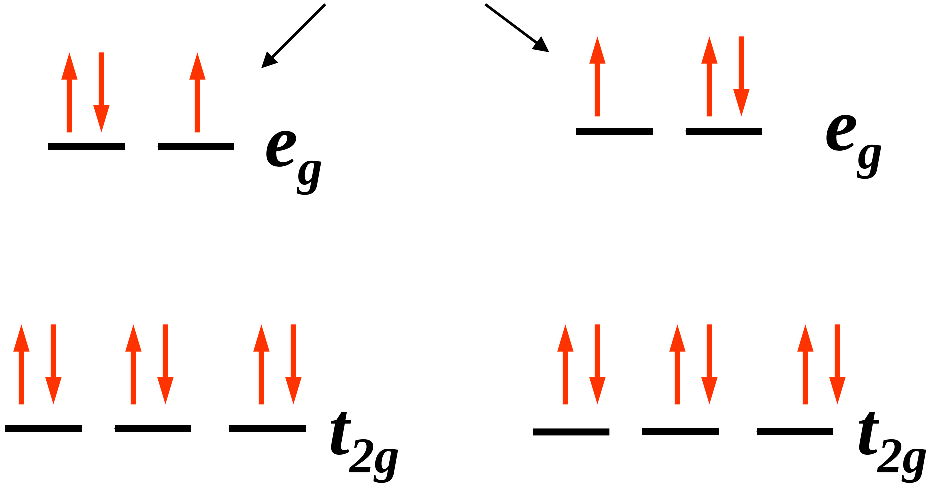
High-spin Ni(II) – only one way of filling the e_g level

Cu(II) – two ways of filling the e_g level – it is electronically degenerate

Ni(II) d^8



Cu(II) d^9



What is the consequence of ELECTRONIC DEGENERACY ?

Jahn-Teller Distortion in d^9 Complexes

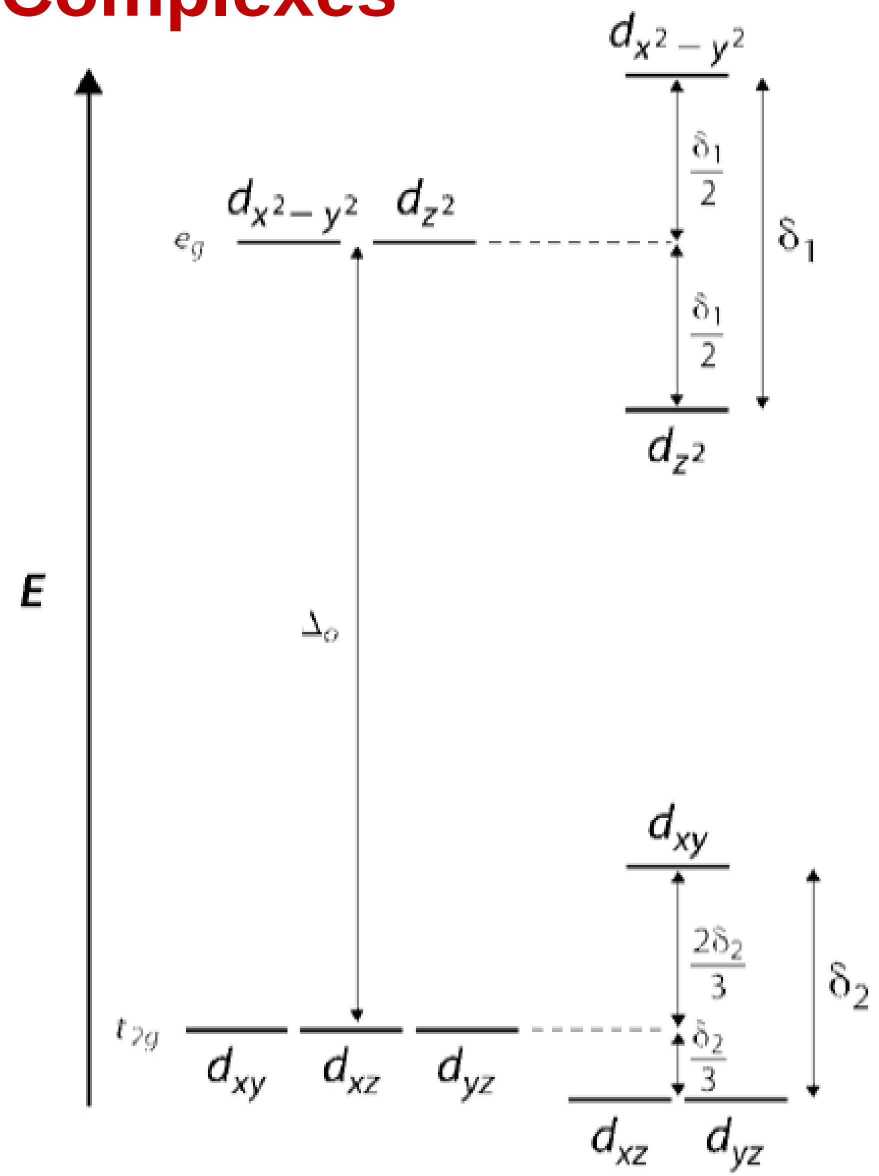
That principle states that *if a system has unequally populated degenerate orbitals, the system will distort to remove the degeneracy.* When the degeneracy is removed, the state of lower energy will be populated first.

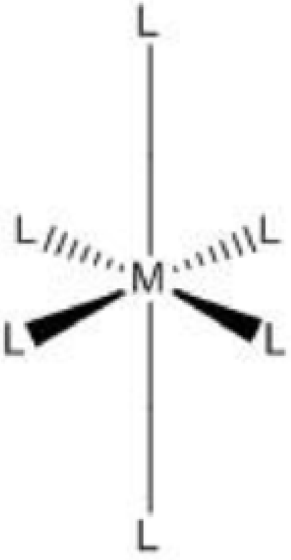
Before Jahn-Teller Distortion

$- 0.4\Delta_0 \times 6 + 3 \times 0.6\,\Delta_0 = -0.6\,\Delta_0$

After Jahn-Teller Distortion

CFSE = $-(0.6\,\Delta_0 + \delta_1/2)$

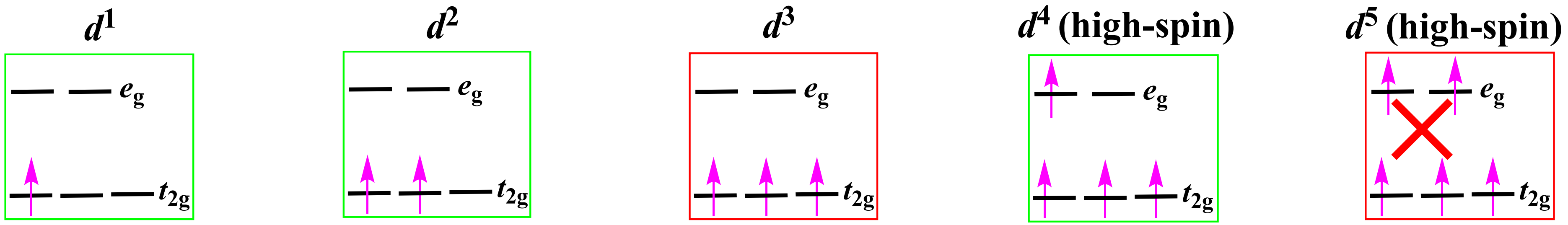


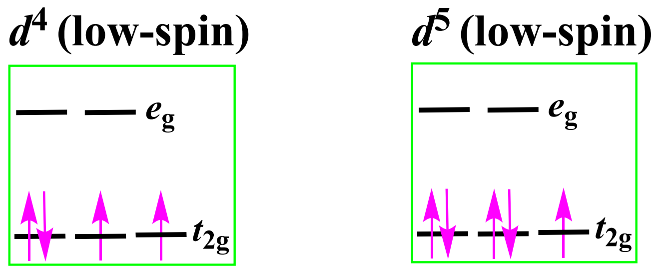


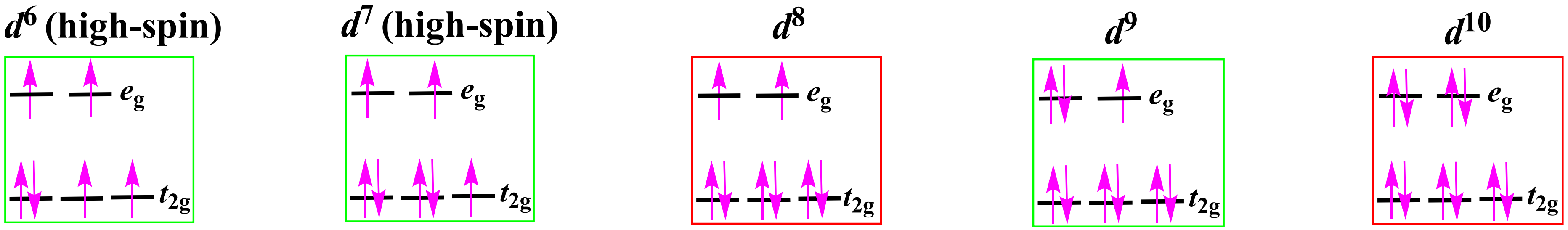
Elongated

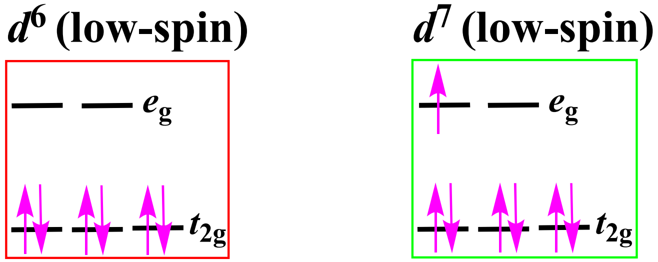
$\Delta o >> \delta 1 > \delta 2.$

Jahn-Teller Distortion



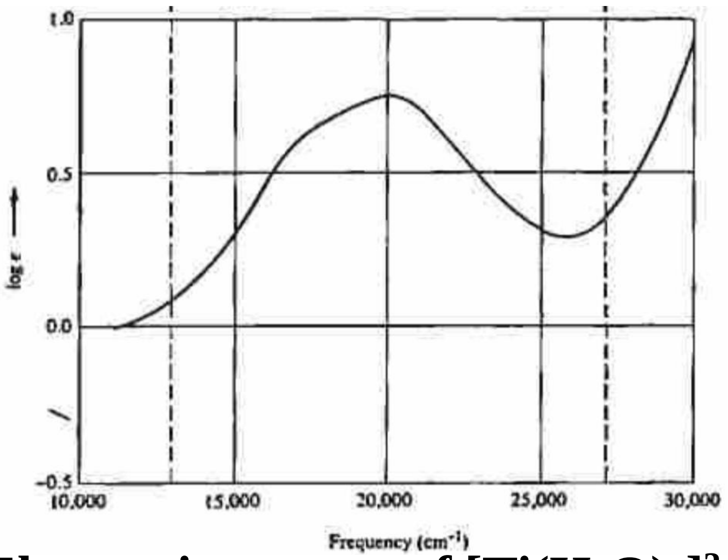




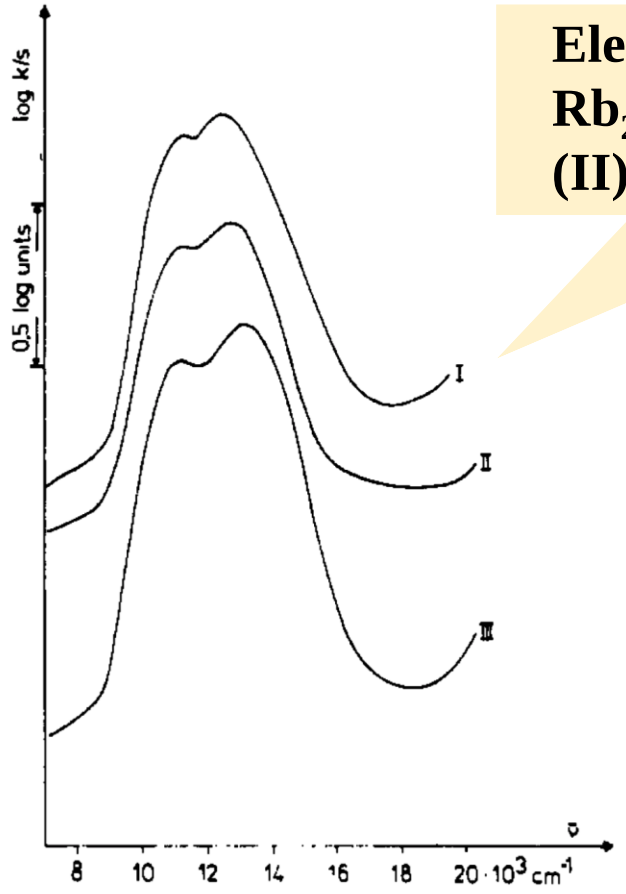


Configurations for which Jahn−Teller distortions are expected in ML_6 complexes

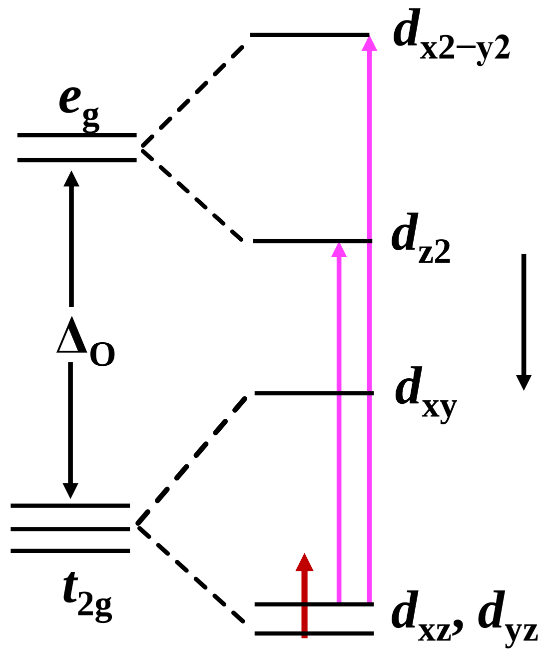
Jahn-Teller Distortion



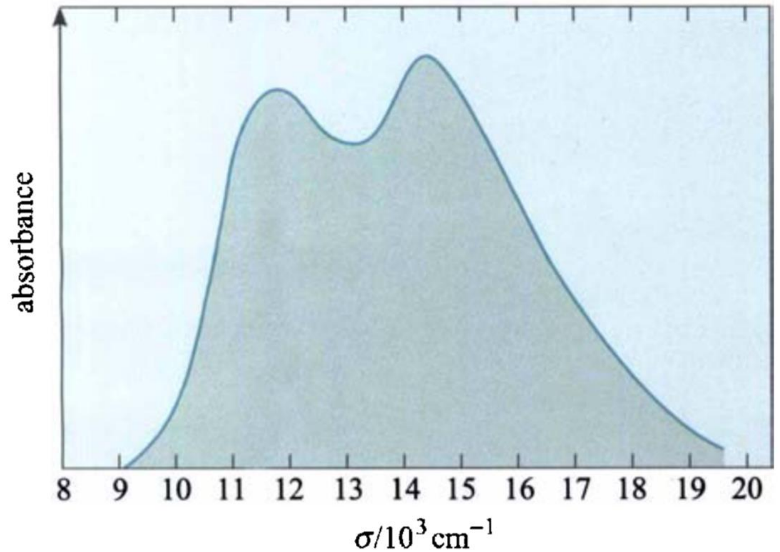
Electronic spectra of $[Ti(H_2O)_6]^{3+}$



Electronic spectra of $Rb_2Na[TiCl_6]$ (I), $Cs_2K[TiCl_6]$ (II), $Rb_3[TiCl_6]$ (III)



Splitting of d-orbitals for Ti^{3+} ion



The electronic absorption spectrum of the $[CoF_6]^{3-}$ ion, showing the two peaks due to the Jahn-Teller splitting of the excited state

Problem solving! (Home work)

$[Co(en)_3]^{2+}$, $[Fe(CN)_6]^{4-}$, $[Fe(CN)_6]^{3-}$, $[CrF_6]^{4-}$

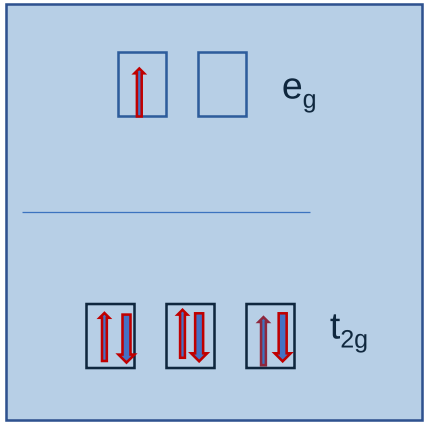
Classify the above given octahedral complexes and write in the boxes below as those

Having no tetragonal distortion	Having slight tetragonal distortion which is seen from UV Visible spectral studies	Having significant tetragonal distortion indicated as varying bond distances in their structure

$[Co(en)_3]^{2+}$ $Co^{2+} = 3d^7$

en = strong field

Electronic degeneracy in the e_g orbital = Significant Jahn Teller distortion



Fe^{II} (LS): No distortion

FeIII (LS): have slight distortion

Cr(III) (HS): will have significant distortion